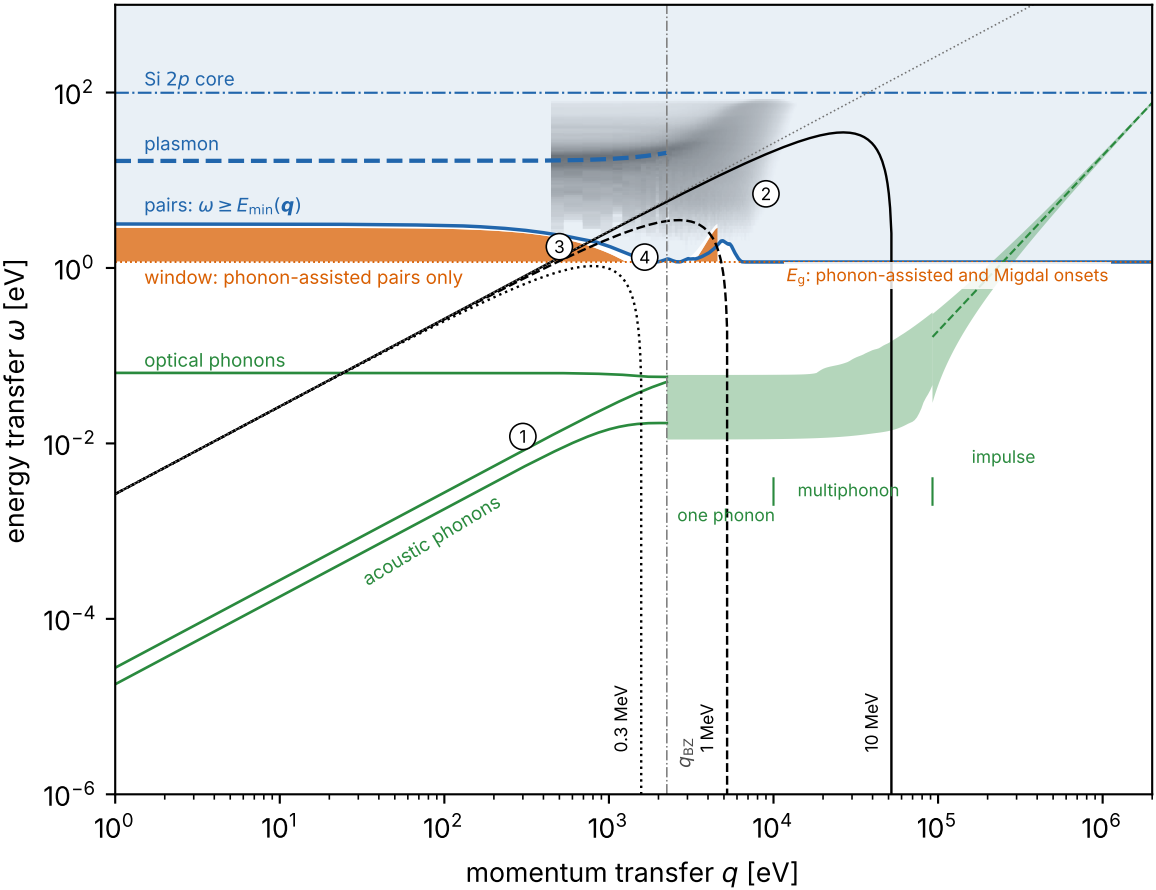

Si 2p core
plasmon
pairs: ω ≥ E_min(q)
window: phonon-assisted pairs only
E_g: phonon-assisted and Migdal onsets
optical phonons
acoustic phonons
one phonon
multiphonon
impulse
0.3 MeV
q_BZ
1 MeV
10 MeV
energy transfer ω [eV]
momentum transfer q [eV]